National University
Of Computer & Emerging Sciences Peshawar Campus

Lab 08
IPv4 Subnetting

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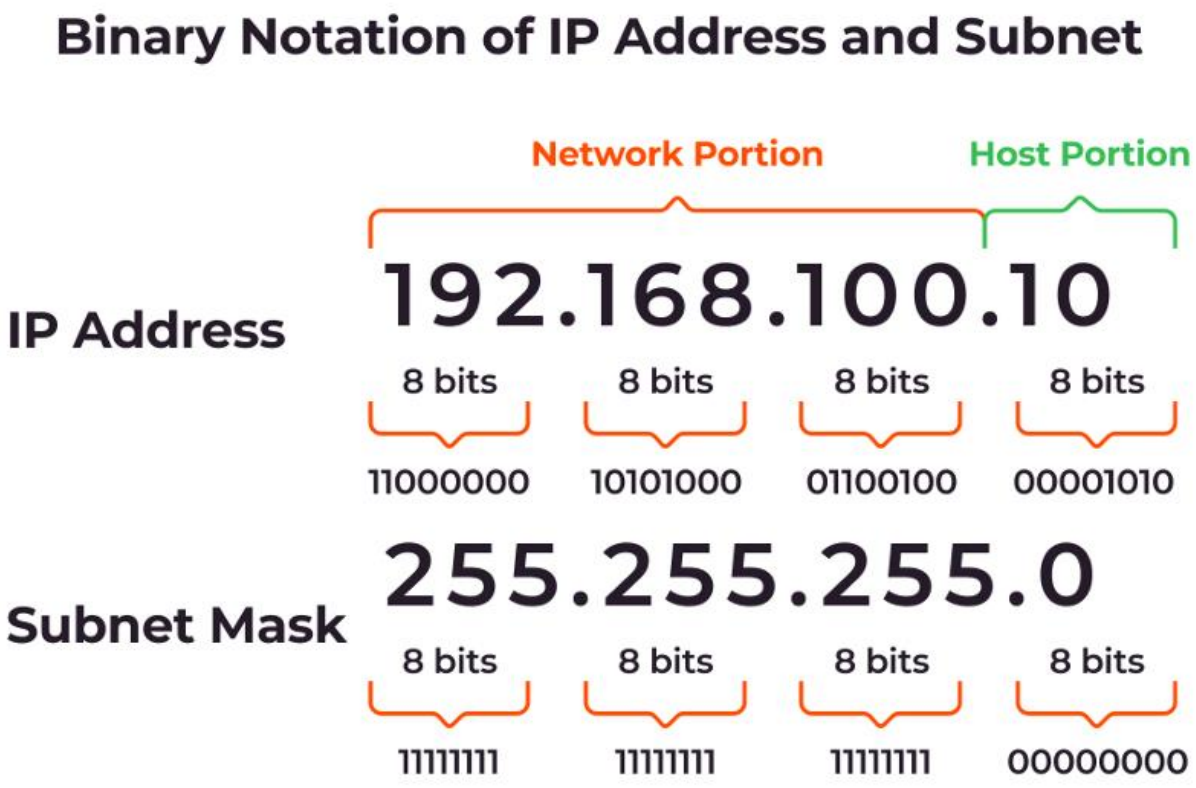


Rule 1

• If the host bits in a given IP address are all set to '0', this is the network or subnet address.
• If the host bits in a given IP address are all set to '1', this is the broadcast address (all hosts in the subnet/network are destination).

Rule 2

$$\text{Number of Subnets} = 2^{s}$$

where 's' = number of subnet bits

- The formula used to calculate the number of available subnets given the specific length of network mask.

Rule 03

$$\text{Number of Hosts} = 2^h - 2$$

where 'h' = number of host bits (unmasked)

- The formula used to calculate the number of available hosts per subnet or network given the specific network mask.

Classful Address Table

Class	First Byte Range	Default Netmask	Default Netmask (binary)
A	1-126	/8 = 255.0.0.0	11111111.00000000.00000000.00000000
B	128-191	/16 = 255.255.0.0	11111111.11111111.00000000.00000000
C	192-223	/24 = 255.255.255.0	11111111.11111111.11111111.00000000

Network (classful) Addresses and Subnet Addresses (classless)

Examples of Network Addresses (Classful)

10.0.0.0/8
172.16.0.0/16
192.168.1.0/24

Examples of Subnet Addresses (Classless)

10.1.0.0/16
172.16.1.0/24
192.168.1.0/28

Number of Subnet Bits (Example)

Example: How many subnets given the prefix?
Address = 192.168.1.0/28

	192	168	1	0
Address	11000000.	10101000.	00000001.	00000000
Netmask	11111111.	11111111.	11111111.	11110000

Default Network Mask
Extended Bits
Host Bits

Number of Subnets Available - Calculation

	192	168	1	0
Address	11000000.	10101000.	00000001.	00000000

Default Network Mask = 24 bits for class C

Subnet Bits

Answer:

Number of Subnets = $2^s = 2^4$ = 16 subnets

Number of Hosts Available - Calculation

	192	168	1	0
Address	11000000.	10101000.	00000001.	00000000

Default Network Mask = 24 bits for class C

Subnet Bits

Host Bits

Answer:

Number of Host Addresses = $2^h-2 = 2^4-2$ = 16 - 2 = 14

Subnet Calculation Aid

Base^{exp}	2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0
Weight	128	64	32	16	8	4	2	1

128
192
224
240
248
252
254
255

Example of Subnet Binary-to-Decimal Conversion

Subnet Netmask Example: /28
24 bits in decimal are: 255.255.255
4 bits in decimal are: 240
Whole Netmask /28 = 255.255.255.240

Base^{exp}	2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0
Weight	128	64	32	16	8	4	2	1
	128	192	224	240	248	252	254	255
	1	1	1	1	0	0	0	0

Question 1

Given the prefix 192.168.1.0/24, what should be the length of subnet mask allowing up to 9 subnets?

Answer 1

- The address belongs to the class C and uses its default network mask.
- That leaves us with 8 bits to play with (the last byte).
- Before we change anything, our address and network mask converted into the binary notation look like shown in the pic.

192.168.1.0/24 in Binary

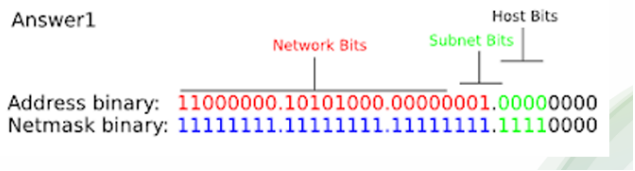
Question1

	Network Bits	Host Bits
Address binary:	11000000.10101000.00000001	.00000000
Netmask binary:	11111111.11111111.11111111	.00000000

Answer 1 (Cont'd)

- In order to create 9 subnets we must extend the existing length of the network mask by 4 bits which allows up to 16 subnets.
- If I tried to extend it by 3 bits only, the maximum subnets allowed would be only 8 subnets.
- So, I must use 4 bits and the result is: 192.168.1.0/28 (192.168.1.0 255.255.255.240).

The Answer to Question 1



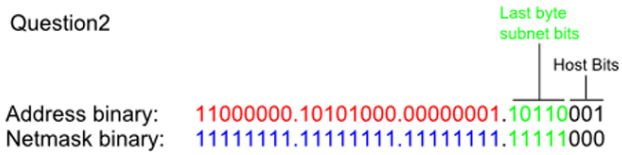
Question 02

Given the host address 192.168.1.177/29, what are the subnet and broadcast addresses?

Answer 2

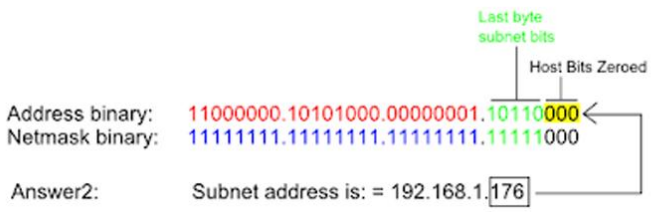
- In order to determine the subnet and broadcast address of the subnet of this host address, we must look at the length of the network mask first. It is 29 bits (24+5).
- This tells us that the last byte of the address has 5 bits masked (subnet bits) and 3 bits unmasked (host bits).
- It is a good idea to look at the last byte of the address (177) with its network mask using binary notation.

192.168.1.177/29 in Binary



Answer 2 (Cont'd)

Host Bits Zeroed = Subnet Address

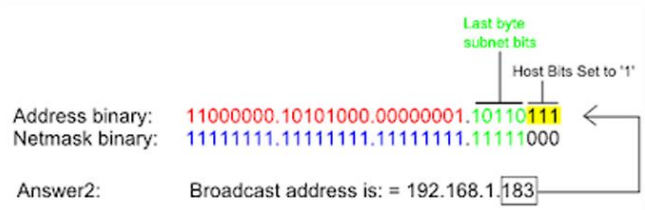


- Since we must determine the the subnet in which the host resides (177 = 10110001), the host portion of the prefix (host bits reside in the last byte) must all be set to '0'.
- The byte value with the host zeroed is the address of the subnet.

Answer 2 (Cont'd)

- The second part of the question relates to the broadcast address of the subnet.
- As you remember, in order to obtain the broadcast address, you must put '1' on all host bits of the subnet/network.
- The subnet has already been determined, so let's put '1' on all bits of the host portion:
  - .10110111
  - .10110000 **= 176 <- subnet address**
  - .00000111 **= 7 <- host bits set to '1'**
- In decimal it is: 176 + 7 = **183**
- The broadcast address is: **183**.

Host Bits Set to '1' = Broadcast Address

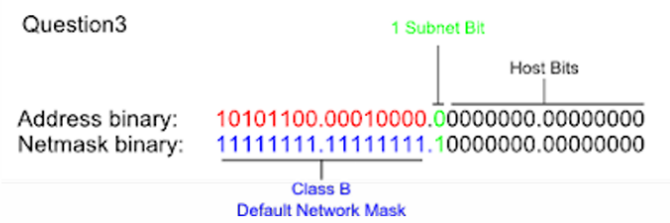


Question 3
Given the prefix 172.16.0.0/17, how many subnets can you create?

Answer 3

- This is a bit tricky isn't it?
- In order to answer this question, you don't need any calculator, paper or pen.
- You must trust the rule 2.
- The address and its network mask (called prefix) converted into binary look like presented here:

The Number of Subnets for 172.16.0.0/17



Answer 3 (Cont'd)

- As you see the number of bits we have extended the class B address is: 1. So, the number of subnets we can create with it is: 2 subnets, since this subnet bit can be either 1 or 0.

Questions 3 Answer

Number of Subnets = $2^1 = 2$

Class A

- **Range**: 0.0.0.0 to 127.255.255.255
- **Default Subnet Mask**: 255.0.0.0 (8 bits for the network, 24 bits for hosts)
- **Purpose**: Used for very large networks (e.g., large organizations, ISPs).
- **Number of Networks**: 128 (2^7, as 0 is reserved).
- **Hosts per Network**: ~16.7 million (2^24 - 2).

Class B

- **Range**: 128.0.0.0 to 191.255.255.255
- **Default Subnet Mask**: 255.255.0.0 (16 bits for the network, 16 bits for hosts)
- **Purpose**: Medium-sized networks (e.g., universities, organizations).
- **Number of Networks**: ~16,384 (2^14).
- **Hosts per Network**: ~65,534 (2^16 - 2).

Class C

- **Range**: 192.0.0.0 to 223.255.255.255
- **Default Subnet Mask**: 255.255.255.0 (24 bits for the network, 8 bits for hosts)
- **Purpose**: Small networks (e.g., small businesses).
- **Number of Networks**: ~2.1 million (2^21).
- **Hosts per Network**: 254 (2^8 - 2).